



YEAR 11 TERM 4 EXAMINATION
2004

EXTENSION MATHEMATICS

Time allowed: 1 hour

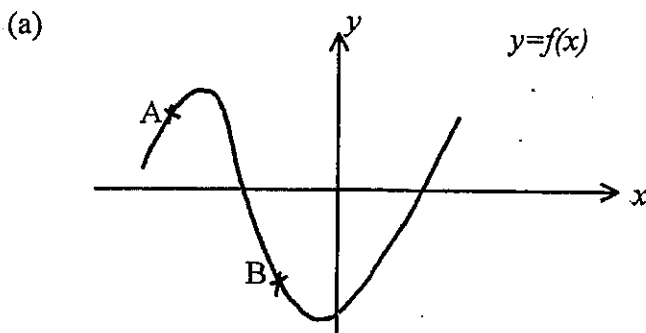
Total marks: 45

All necessary working should be shown.

START EACH QUESTION ON A NEW PAGE.

Question 1

Marks



The graph of $y=f(x)$ is shown. Determine the sign of -

- | | | |
|------|---|---|
| (i) | $f'(x)$ at A | 1 |
| (ii) | $f''(x)$ at B | 1 |
| (b) | Find a primitive function of $3x^2 - \frac{1}{3x^2}$ | 2 |
| (c) | The gradient at any point (x, y) on a curve is $3x^2 - 2x + 3$. Find the equation of the curve given that it passes through the point $(3, 3)$. | 2 |

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Question 2**Marks**

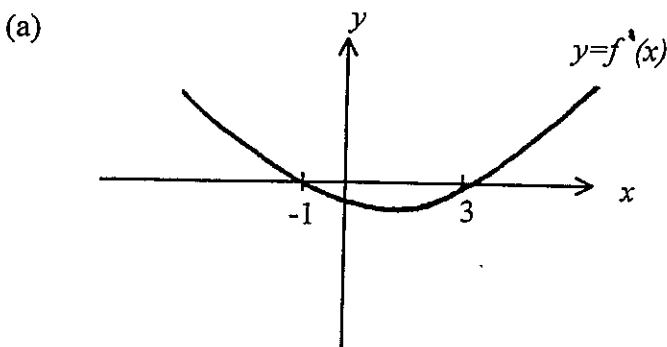
For the curve $y = \frac{x^2 - 1}{x^2 - 4}$

- (a) Find any stationary points and determine their nature. 3
- (b) Determine the equations of any (i) vertical asymptotes
(ii) horizontal asymptotes 3
- (c) Draw a neat sketch of the curve showing its main features, including the intercepts with the axes. 3
- (d) Find the greatest value of the function in the domain $-3 \leq x \leq 3$ 1

Question 3

- (a) Show that there is a point of inflection at (2, 6) on the curve $y = x^3 - 6x^2 + 8x + 6$ 3
- (b) For a certain function $f'(x) = \frac{(x+3)(x-1)^{16}}{\sqrt{x^2 - 4x + 5}}$ 2

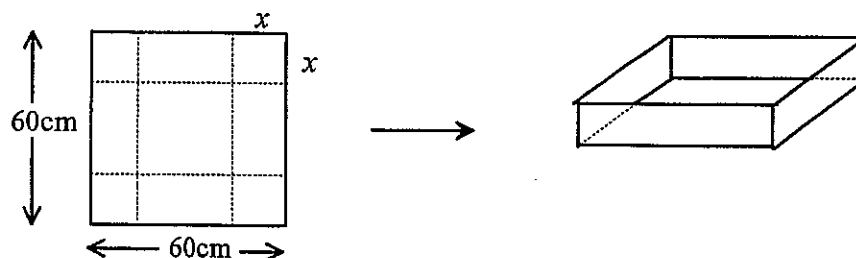
Determine the nature of the stationary point at $x = 1$.

Question 4

The diagram shows the graph of the derived function $y = f'(x)$ for a given function $y = f(x)$.

Copy the diagram and on it show clearly a possible graph of $y = f(x)$. 3

(b)



A square sheet of metal of side 60cm has squares of side x cm cut from each corner, as shown above. The edges are then bent up to form an open tray.

- (i) Show that the volume of the tray is $V = 3600x - 240x^2 + 4x^3$ 1
- (ii) Hence find the value of x which makes this volume a maximum. 3

Question 5

- (a) (i) How much does \$1200 amount to if it is invested for 20 years at 9% p.a. interest compounded monthly? 1
- (ii) Harry contributes \$1200, at the beginning of every year, into a superannuation fund which pays 9% p.a. interest compounded monthly. How much will be in the fund at the end of the twentieth (20th) year? 3
- (b) Sue borrows \$10,000 from a bank which charges 12% p.a. interest on the balance owing at the end of each month. She agrees to repay the loan in 36 equal monthly repayments.
- (i) Find the amount of each monthly repayment. 3
- (ii) Calculate the amount still owing after 1 year, (i.e. after the 12th repayment) 2
- (iii) How much interest was charged for the first year? 2

Question 6

Prove by mathematical induction that $n^3 - n$ is a multiple of 6 for all positive integers $n \geq 2$. 6

END OF PAPER

Q1
a) (i) +
(ii) [0 if pt. of inflection]
but concave up $\therefore f''(x) > 0$

b) $f'(x) = 3x^2 - \frac{x^{-2}}{3}$

$f(x) = x^3 - \frac{1}{3} \frac{x^{-1}}{-1} + C$

$= x^3 + \frac{1}{3x} + C$

c) $y' = 3x^2 - 2x + 3$

$y = x^3 - x^2 + 3x + C$

but $x=3, y=3$

$3 = 27 - 9 + 9 + C$

$C = -24$

$y = x^3 - x^2 + 3x - 24$

Q2. $y = \frac{x^2-1}{x^2-4}$

a) $y' = \frac{(x^2-4)(2x) - (x^2-1)(2x)}{(x^2-4)^2}$

$= \frac{2x(x^2-4-x^2+1)}{(x^2-4)^2}$

$= \frac{-6x}{(x^2-4)^2}$

for st. pts $y' = 0$

$-6x = 0 \Rightarrow x = 0$

x	-1	0	1
y'	+	0	-

$\therefore$ max at $(0, \frac{1}{4})$

for vertical asymptotes

$x^2 - 4 = 0$
 $x = \pm 2$

for horizontal asymptotes.

$\lim_{x \rightarrow \pm\infty} \frac{x^2-1}{x^2-4} = \lim_{x \rightarrow \pm\infty} \frac{1 - \frac{1}{x^2}}{1 - \frac{4}{x^2}}$

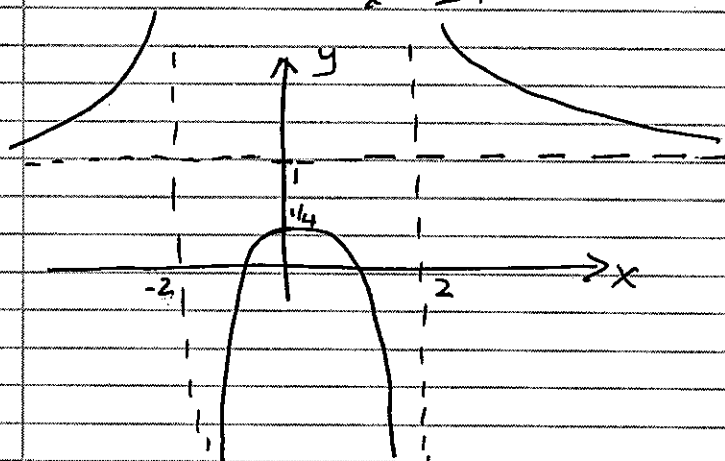
$= 1$

$\frac{1 - \frac{1}{100}}{1 - \frac{4}{100}} > 1$

c) $x=0, y = \frac{1}{4}$

note $-2 < x < 2, y \leq \frac{1}{4}$

x intercepts $x^2 - 1 = 0$
 $x = \pm 1$



for ~~$x \leq -3$~~ $-3 \leq x \leq 3$

max value of function

~~at $x = \pm 3$~~
 ~~$y = \frac{9-1}{9-4} = \frac{8}{5}$~~

There is no max value.

as $x \rightarrow -2^-, y \rightarrow +\infty$
 $x \rightarrow 2^+, y \rightarrow +\infty$

Q3.

$$y = x^3 - 6x^2 + 8x + 6$$

$$y' = 3x^2 - 12x + 8$$

$$y'' = 6x - 12$$

for inflection $y'' = 0$

$$6x - 12 = 0$$

$$x = 2$$

note at $x = 2$

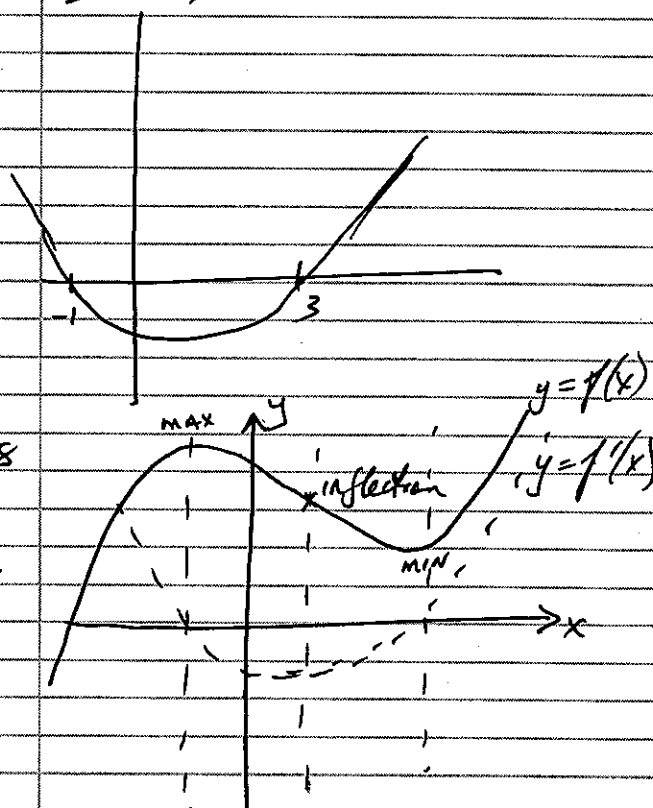
$$y' = 12 - 24 + 8 > 0$$

 $\therefore$ not a max/min

$$\text{if } x = 2 \quad y = 8 - 24 + 16 + 6 = 6$$

 $\therefore (2, 6)$ is a pt. of inflection

Q4 a)



$$b) \quad f'(x) = \frac{(x+3)(x-1)}{\sqrt{x^2-4x+5}}$$

possible st. points at -3 and 1.

x	-4	-3	0	1	2
f'(x)	-	0	+	0	+

\ — / = /

at $x = 1$ there is a horizontal pt. of inflection.

b)

(i)

x		x
	$60 - 2x$	

note $x < 30$

$$V = (60 - 2x)^2 \cdot x$$

$$= (3600 - 240x + 4x^2)x$$

$$V = 3600x - 240x^2 + 4x^3$$

$$(ii) \text{ for max/min } \frac{dV}{dx} = 0.$$

$$V' = 3600 - 480x + 12x^2$$

$$= 12(300 - 40x + x^2)$$

$$= 12(x^2 - 40x + 300)$$

$$= 12(x - 10)(x - 30) \quad \text{but } x < 30$$

$$= 0 \quad \text{if } x = 10$$

$$V'' = -480 + 24x$$

$$\text{if } x = 10 \quad V'' = -240 < 0$$

$\therefore$ max volume if $x = 10$

Q5 20 years = 240 months.

a) 1) $9\% \text{ pa} = \frac{9}{12}\% \text{ p/month}$
 $= 0.75\%$

$$\begin{aligned} \text{Amount} &= P \left(1 + \frac{r}{100}\right)^n \\ &= 1200 (1.0075)^{240} \\ &= 7210.981829 \\ &= \$7210.98 \end{aligned}$$

(ii) Total = $1200(1.0075)^{240}$
 $+ 1200(1.0075)^{228}$
 $+ 1200(1.0075)^{216}$
 $\vdots$
 $+ 1200(1.0075)^{12}$ } 20 terms

$$\begin{aligned} \text{GP } a &= 1200(1.0075)^{12} \\ r &= (1.0075)^{12} \quad n = 20. \end{aligned}$$

$$\begin{aligned} S_{20} &= \frac{a(r^{20} - 1)}{r - 1} \\ &= 1200(1.0075)^{12} \left[\frac{(1.0075)^{240} - 1}{(1.0075)^{12} - 1} \right] \end{aligned}$$

$$= \$70089.23171 \dots$$

$$= \$70089.23$$

Q5b. $M = \text{payment/monthly}$

$A_n = \text{money due at end of month } n$

$$12\% \text{ pa} = 1\% \text{ p/month}$$

$$A_1 = 10000 \times 1.01$$

$$A_2 = (10000 \times 1.01 - M) \times 1.01$$

$$A_3 = (10000 \times 1.01^2 - (M \times 1.01)) \times 1.01$$

$$= 10000 \times 1.01^3 - M \times 1.01^2$$

$$A_4 = 10^4 \times 1.01^4 - M \times 1.01^3 - M \times 1.01$$

$$A_5 = 10^4 \times 1.01^5 - M \times 1.01^4 - M \times 1.01^2$$

something is wrong.

start again

$M = \text{monthly payment}$

$A = \text{Money due after monthly payment}$

$$A_1 = (10000 - M) \times 1.01$$

$$= 10^4 \times 1.01 - M$$

$$A_2 = (A_1) \times 1.01 - M$$

$$= 10^4 \times 1.01^2 - M \times 1.01 - M$$

$$A_3 = A_2 \times 1.01 - M$$

$$= 10^4 \times 1.01^3 - M(1.01^2 + 1.01 + 1)$$

$$A_{36} = 10^4 \times 1.01^{36} - M(1.01^{35} + \dots + 1)$$

$$\text{but } A_{36} = 0$$

$$M = \frac{10^4 \times 1.01^{36}}{1 + 1.01 + 1.01^2 + \dots + 1.01^{35}}$$

$$\text{GP } a = 1, r = 1.01$$

$$n = 36.$$

$$= \frac{10^4 \times 1.01^{36}}{\left[\frac{1 \times 1.01^{36} - 1}{0.01} \right]} = \frac{10^4 \times 1.01^{36}}{(1.01^{36} - 1)}$$

$$M = 332.143098 \dots$$

$$\text{Monthly repayment} = \$332.14$$

5b(ii)

$$A_{12} = 10^4 \times 1.01^{12} - M(1 + 1.01 + 1.01^2 + \dots + 1.01^{11})$$

$$= 10^4 \times 1.01^{12} - 332.14 \left(\frac{1 \times (1.01^{12} - 1)}{0.01} \right)$$

$$= 7055.88375 \dots$$

$\therefore 7055.88$ still owing after 1 year.

5b(iii)

~~$$10^4 - 7055.88 = 2944.12$$~~
~~$$(12 \times 332.14)$$~~

10^4 borrowed

7055.88 owing

2944.12 paid off

but 332.14×12 paid

$$= 3985.68$$

$$\begin{array}{r} \text{Interest} = 3985.68 \\ 2944.12 \\ \hline 1041.56 \end{array}$$

$$= \$1041.56$$

Q6 to show $n^3 - n \div 6 \quad \forall n \geq 2$
P(n)

$\nexists n=2 \quad 8-2=6$ - a multiple of 6

so P(n) true for $n=2$.

~~assume true for~~

P(k)

$$k^3 - k = 6m$$

Assume true P(k) true

P(k+1)

$$(k+1)^3 - (k+1) = (k+1)((k+1)^2 - 1)$$

$$= (k+1)(k^2 + 2k)$$

$$= k^3 + 2k^2 + k^2 + 2k$$

$$= k^3 - k + 3k^2 + 3k$$

$$= 6m + 3[k(k+1)]$$

must have
a factor
of 2 = 2p.

$$= 6m + 6p$$

$$= 6(\dots)$$

$\therefore$ has a factor of 6

$$P(k) \text{ true} \Rightarrow P(k+1) \text{ true}$$

~~by~~

True for $n=2 \therefore$ true $n=3$

$n=3 \therefore$ true $n=4$

by PMI true for all $n \geq 2$.